

Software Design Document (SDD)

Module	OLED Driver (SSD1306)
Version	1.0.0
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1. Introduction

1.1. Purpose

This document describes the design and architecture of the OLED driver module for the STM32F411RE Nucleo board, as used in the MSE Digital Oscilloscope capstone project. It covers the register-level initialization sequence of the SSD1306 controller, the frame-buffer model, and the full public API. The intended audience is any developer who needs to integrate, extend, or test the driver.

1.2. Scope

The OLED driver is a low-level embedded software component that abstracts the SSD1306 OLED controller on the GME12864-11 display module (128x64 px, I2C address 0x3C). It depends on the I2C driver (i2c.h / i2c.c) for all bus transactions and exposes a typed C API for initialization, pixel manipulation, waveform rendering, status-bar drawing, and display flushing. The scope does not include the I2C peripheral configuration (see SDD_I2C_Driver).

1.3. Definitions and Acronyms

Term	Definition
SSD1306	Solomon Systech controller IC driving the OLED panel.
Frame buffer	1024-byte SRAM array (8 pages x 128 columns) mirroring display RAM.

Page	Horizontal band of 8 pixel rows; SSD1306 has 8 pages (0-7).
I2C	Inter-Integrated Circuit serial bus at 400 kHz (Fast Mode).
ADC	Analog-to-Digital Converter; 12-bit on STM32F411RE (0-4095).
CH1 / CH2	Oscilloscope channels 1 and 2, mapped to ADC inputs PA0 and PA1.
V/div	Volts per grid division; depends on the amplitude zoom setting.

2. Driver Overview

2.1. Hardware Configuration

The GME12864-11 OLED module uses the SSD1306 controller in Page Addressing Mode over I2C. The 7-bit I2C address is 0x3C (SA0 pin tied to GND). Supply voltage is 3.3 V; the built-in charge pump generates the panel drive voltage internally.

Parameter	Value
Resolution	128 x 64 pixels
I2C address	0x3C (7-bit)
I2C speed	400 kHz (Fast Mode)
Supply	3.3 V
Frame buffer size	1024 bytes (8 pages x 128 columns)
Addressing mode	Page Addressing (0x02)
Font	5x7 bitmap (95 ASCII chars, 32-126)

2.2. Frame Buffer Model

The driver maintains a static 1024-byte frame buffer. Each byte represents 8 vertically stacked pixels; bit 0 is the topmost pixel of the page and bit 7 is the bottommost. To modify pixel (x, y):

```
page = y / 8
```

```
bit = y % 8
```

```
frame_buf[page * 128 + x] |= (1 << bit); // set ON
```

```
frame_buf[page * 128 + x] &= ~(1 << bit); // set OFF
```

No changes are visible on the display until `oled_flush()` is called. This separation allows multiple drawing primitives to compose a full frame before committing it via a

single I2C burst per page.

2.3. Display Layout

The 64-pixel height is partitioned into three regions:

- * Status bar top (y = 0-7): V/div, timebase, frequency, and channel labels.
- * Waveform area (y = 8-55): oscilloscope grid and signal traces.
- * Status bar bottom (y = 56-63): real-time voltage readout for CH1 and CH2.

3. API Reference

3.1. Data Types

Type	Values	Description
Oled_Status_t	OLED_OK = 0 OLED_ERROR = 1	Return code for functions that perform I2C transactions.
Oled_Color_t	OLED_COLOR_BLACK = 0 OLED_COLOR_WHITE = 1	Pixel color: WHITE turns pixel on; BLACK turns it off.

3.2. Constants

Constant	Value	Description
OLED_WIDTH	128	Display width in pixels.
OLED_HEIGHT	64	Display height in pixels.
OLED_PAGES	8	Number of SSD1306 pages (HEIGHT / 8).
OLED_ADDR	0x3C	7-bit I2C address (SA0 = GND).
OLED_ADC_MAX	4095	Full-scale 12-bit ADC value.

3.3. Function Reference

`Oled_Status_t oled_init(void)`

Sends the SSD1306 initialization command sequence (display off, clock, multiplex, charge pump, page addressing mode, segment remap, COM scan direction, contrast, precharge, VCOMH, display-follows-RAM, normal polarity, display on). Calls `oled_clear()` to zero the frame buffer. Must be called after `i2c_init()`.

Returns: OLED_OK on success; OLED_ERROR if any I2C transaction fails.

```
void oled_clear(void)
```

Zeroes all 1024 bytes of the frame buffer (turns off every pixel in software). The display is not updated until `oled_flush()` is called.

```
void oled_set_pixel(uint8_t x, uint8_t y, Oled_Color_t color)
```

Sets or clears one pixel in the frame buffer. Out-of-range coordinates are silently ignored. Does not flush to the display.

Parameters:

- * x -- column index, 0-127.
- * y -- row index, 0 (top) to 63 (bottom).
- * color -- `OLED_COLOR_WHITE` or `OLED_COLOR_BLACK`.

```
void oled_draw_grid(void)
```

Draws the oscilloscope graticule: solid separator lines between the status bars and the waveform area, left and right border lines, dot-matrix grid ticks every 16 px horizontally and vertically, and a dashed center line at $y = 32$ (0 V reference).

```
void oled_draw_waveform(const uint16_t *samples, uint16_t n_samples)
```

Renders CH1 as a solid connected trace. Each display column ($x = 1-126$) maps to the corresponding sample; the ADC value is mapped to the waveform area ($y = 9-54$) with the current amplitude zoom. Adjacent pixels are connected vertically to avoid gaps.

Parameters:

- * samples -- pointer to ADC sample array (values 0-4095).
- * n_samples -- number of samples (ideally 128).

```
void oled_draw_waveform_ch2(const uint16_t *samples, uint16_t n_samples)
```

Renders CH2 as a dotted trace (odd columns only) to distinguish it from CH1. The ADC value is mapped symmetrically around 2048 (0 V) using the current amplitude zoom.

Parameters:

- * samples -- pointer to ADC sample array (values 0-4095).
- * n_samples -- number of samples (ideally 128).

```
void oled_set_amplitude_scale(uint16_t scale)
```

Sets the global amplitude zoom factor used by both waveform drawing functions. Values outside [256, 4095] are clamped.

Parameters:

- * scale -- ADC range visible on screen, centered at 2048. Range: 256-4095. 4095 = no zoom (1x); 2048 = 2x; 1024 = 4x; 256 = 16x.

```
void oled_draw_circle(uint8_t cx, uint8_t cy, uint8_t r, Oled_Color_t color)
```

Draws the outline of a circle using the Bresenham midpoint algorithm. Pixels that fall outside display bounds are silently clipped.

Parameters:

- * cx -- center column (0-127).
- * cy -- center row (0-63).
- * r -- radius in pixels.
- * color -- OLED_COLOR_WHITE or OLED_COLOR_BLACK.

```
void oled_draw_char(uint8_t x, uint8_t y, char c)
```

Blits a 5x7 glyph from FONT5X7 into the frame buffer using bitwise OR. Characters outside the printable range are rendered as a space.

Parameters:

- * x -- column of the leftmost pixel (0-122).
- * y -- row; must be a page boundary (0, 8, 16, 24, 32, 40, 48, or 56).
- * c -- ASCII character (32-126).

```
void oled_draw_string(uint8_t x, uint8_t y, const char *str)
```

Calls oled_draw_char() for each character in str, advancing x by 6 pixels (5 px glyph + 1 px spacing). Stops when the string ends or the right margin is reached.

Parameters:

- * x -- starting column.
- * y -- page-aligned row.
- * str -- null-terminated ASCII string.

```
void oled_draw_statusBars(uint32_t inter_sample_us, uint32_t freq_hz,  
uint16_t pot_amp, uint16_t ch1_adc, uint16_t ch2_adc)
```

Draws both status bars. Top bar: CH1 circle + V/div label, timebase string, frequency string, CH2 circle. Bottom bar: voltage strings for both channels. CH2 voltage compensates for the LM358 op-amp bias ($V_{\text{real}} = 2 * V_{\text{adc}} - 3.3 \text{ V}$).

Parameters:

- * `inter_sample_us` -- time between ADC samples in us; 0 = Auto mode.
- * `freq_hz` -- estimated CH1 frequency in Hz; 0 = no signal detected.
- * `pot_amp` -- raw ADC reading from the amplitude potentiometer (0-4095).
- * `ch1_adc` -- latest ADC sample from CH1 for voltage display.
- * `ch2_adc` -- latest ADC sample from CH2 for voltage display.

`Oled_Status_t oled_flush(void)`

Transmits all 8 pages of the frame buffer to the SSD1306 via I2C. For each page it sends a 3-byte set-page command (0xB0|page, 0x00, 0x10) followed by 129 bytes (control byte 0x40 + 128 data bytes). Total transfer per frame: ~1056 bytes.

Returns: OLED_OK on success; OLED_ERROR if any I2C write fails.

4. Implementation Notes

4.1. Initialization Sequence

The SSD1306 requires a specific power-on command sequence before displaying content. The driver sends 18 commands in a single I2C transaction via `oled_send_cmd()`. Key configuration choices:

- * Page Addressing Mode (0x02): simplest mode; each page is written independently.
- * Segment remap (0xA1) + COM scan decrement (0xC8): flips image for correct physical orientation on GME12864-11.
- * Charge pump enabled (0x8D, 0x14): required for single-supply 3.3 V operation.
- * Contrast 0xCF: near-maximum brightness for indoor readability.

4.2. Waveform Rendering Pipeline

Each oscilloscope frame follows this sequence:

1. `oled_clear()` -- zero frame buffer.
2. `oled_draw_grid()` -- draw graticule.

3. `oled_draw_waveform()` -- render CH1 solid trace.
4. `oled_draw_waveform_ch2()` -- render CH2 dotted trace.
5. `oled_draw_statusBars()` -- overlay text labels.
6. `oled_flush()` -- send frame buffer to display.

4.3. I2C Dependency

All bus transactions use `i2c_write(addr, buf, len)` from the I2C driver. The driver does not call `i2c_init()`; the caller must initialize the I2C peripheral at 400 kHz before calling `oled_init()`. All writes are blocking (no DMA or interrupt mode).

4.4. Memory Footprint

Item	Size
frame_buf (SRAM)	1024 bytes
tx_page_buf (SRAM)	129 bytes
FONT5X7 (Flash)	475 bytes
Code (Flash, approx.)	~1.8 kB

5. Revision History

Version	Date	Author	Description
1.0.0	04/06/2026	Carlos Cayetano Adrian Amaro Adalberto Aguirre Daniel López	Initial release covering the full OLED driver API for the MSE